

THE DESK BUDDY

All in one solution to office health

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Introduction

The Desk Buddy is a way for associates to be healthier in the workplace by looking at posture and standing statistics while keeping the office the most suitable for associates by looking at temperature, humidity and noise levels at the same time. This as a whole device is the Desk Buddy and why I chose to make this for my final year project

Objective

1. **Monitor** sit/stand statistics as well as slouch score to get a read on being healthier in the office
2. **Track** temperature, humidity and noise levels in real time
3. **Classify** sitting vs standing states using smoothed machine logic
4. **Alert** users when thresholds are exceeded (noise, slouch temperature etc...)

Hardware

ESP32 - Micro-controller, serial to JSON output
DHT22 - Temperature & Humidity sensor
HC-SR04 - Ultrasonic sensor for desk height
Microphone - ambient audio levels dB
Webcam - Slouch score statistics

Technologies

ESP32

DHT22

HC-SR04

USB Serial

FastAPI

React + Vite

Python

WebSocket

PostgreSQL

SQLAlchemy

PySerial

Docker

Recharts

SoundDevice

OpenCV

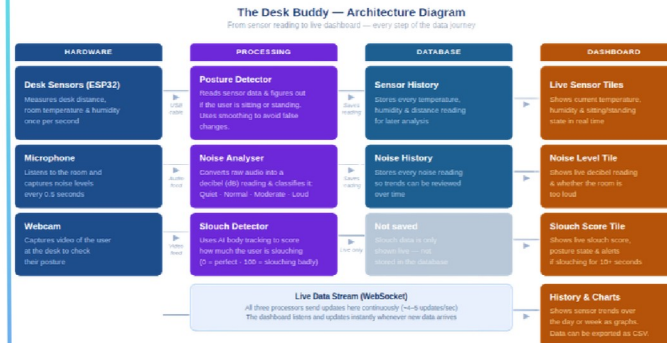
Uvicorn

MediaPipe

Node.js

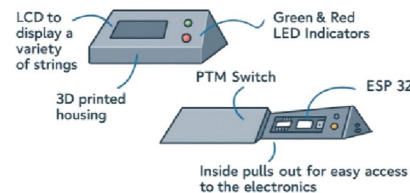
Windows Batch

Architecture Diagram



Project Image

The Desk Buddy



Results

- WebSocket stream delivers ~4-5 messages/sec from 3 parallel sources
- Posture classified as sitting/standing using ± 5 cm hysteresis and 3-second time gate – prevents false transitions
- Slouch confidence scored 0-100 from 7 MediaPipe landmarks; states: Good (0-34) · Warning (35-59) · Slouching (60+)
- ~5 sensor rows/sec written to PostgreSQL – ~18,000 rows per hour of use
- Noise classified across 4 bands: Quiet <50 dB · Normal 50-60 · Moderate 60-70 · Loud ≥ 70 dB
- History view queries PostgreSQL directly via REST and renders as Recharts line charts

How It Works

- ESP32 sensors measure desk distance, temperature, and humidity every second and send a JSON packet over USB
- The serial worker receives the packet, computes a smoothed posture state (sitting/standing) with hysteresis, and forwards data to the backend
- Two parallel workers independently monitor the microphone (noise dB) and webcam (MediaPipe slouch score) at 2 msg/sec each
- FastAPI writes all readings to PostgreSQL and broadcasts every update to the frontend via WebSocket
- The React dashboard merges ~4-5 messages/sec into a single live state – displaying sensor tiles, posture state, and slouch alerts in real time

Conclusion

- The Desk Buddy successfully demonstrates real time office wellness monitoring. It combines hardware and software, vision and audio into a unified system
- Provides feedback to improve your posture comfort and productivity by using The Desk Buddy's software and hardware simultaneously demonstrating a practical low cost alternative to expensive ergonomic tools
- Personalised calibration to adapt the system to each user of the desk buddy rather than a one size fits all which is key to improving accuracy.
- Over a typical session, thousands of readings are stored and the ability to visualise it in a historical trend gives users insights into their habits over time not just in the moment

The Desk Buddy Main Dashboard

